

Section A - True/False with explanation

Table 1 True/False with explanation

No	Answer	Explanation
1	True	Java supports modular programming using methods and classes
2	False	ArrayList is not used to store fixed size elements only.
3	True	The Scanner class is used for user input in console apps.
4	True	A constructor must have a return type in java
5	True	Object-oriented code helps improve reusability and maintainability.
6	False	System.out.println() adds a newline after printing. But for System.out.print don't add a newline.
7	True	ArrayList allows adding duplicate values.
8	False	Overloading only can use for Polymorphism
9	False	You can have more than one class per Java File
10	True	Modular code helps simulate mobile apps in console projects.

Section B

QUESTION 1

```
package Resit_FA_LOWKAHSHENG_SCSJ2500286;

public class Building {

    static String name;

    static String purpose;

    protected Building (String name, String purpose) {

    }

    public static void Dairy (String n, String p)

    {

        name = n;

        purpose = p;}

    public void displayInfo(){

        System.out.println("Name:" + name);

        System.out.println("Purpose:" + purpose);}

    public static void main(String[] args) {

        Building d = new Building (name = "Library", purpose = "Study Area");

        Building e = new Building (name = "Canteen", purpose = "Dining Area");

        d.displayInfo();

        e.displayInfo();

    }

}
```

Figure 1 Corrected code

QUESTION 2

2.1 Tracing Table

Table 2 Tracing table

i	i <= 5	S.O.P	i++	j	j>=0	S.O.P	j--
1	1<=5 (T)	1	1+1=2	3	3>=0 (T)	3	3-1=2
2	2<=5 (T)	2	2+1=3	2	2>=0 (T)	2	2-1=1
3	3<=5 (T)	3	3+1=4	1	1>=0 (T)	1	1-1=0
4	4<=5 (T)	4	4+1=5	0	0>=0 (T)	0	0-1= -1
5	5<=5 (T)	5	5+1=6	-1	-1>=0 (F)	TERMINATE LOOP	
6	6<=5 (F)	TERMINATE LOOP					

2.2

i = 5

j = 0

2.3

totalSteps=5

totalBacktracks=0